From Theory to Practice in Wireless Multimedia Delivery

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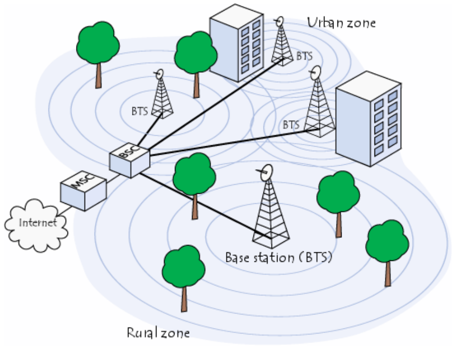
Princeton University SCENIC
Social and Clean Edge Network via Inter-disciplinary Collaboration

The Princeton EDGE Lab

Outline

- Research Spectrum
- Content-Aware Optimization in W-CDMA (EV-DO, LTE)
- QAVA: Quota-Aware Video Adaptation
- Implementation (WARP, Android)
- Conclusions

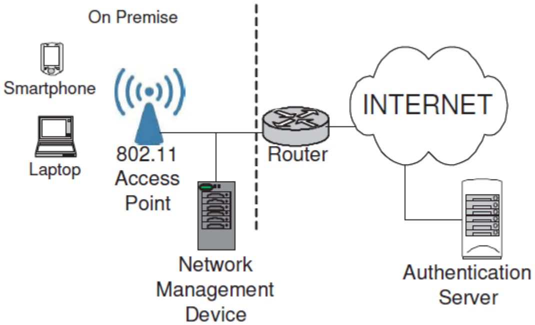
Research Spectrum



Cellular

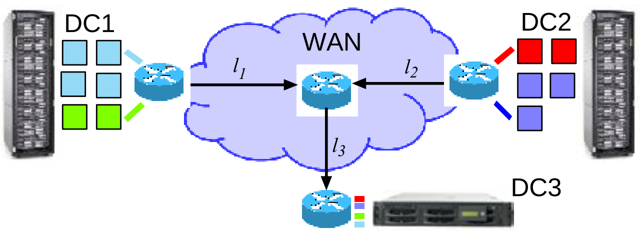
- ❑ Content-aware optimization for wireless video delivery
- ❑ QAVA: Quota-Aware Video Adaptation

Wi-Fi



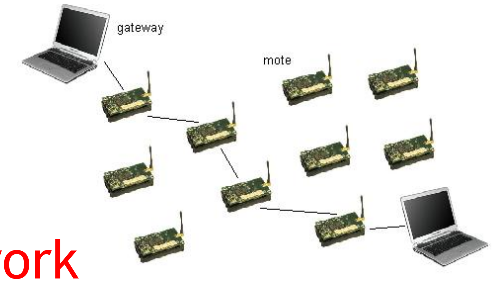
- ❑ Wi-Fi traffic modeling in large-scale public hotspots

- ❑ Multi-class traffic management in data center backbone networks



Data center backbone

- ❑ Throughput-delay tradeoff for fast data collection
- ❑ Optimal routing topologies
- ❑ Interference mgmt: power control

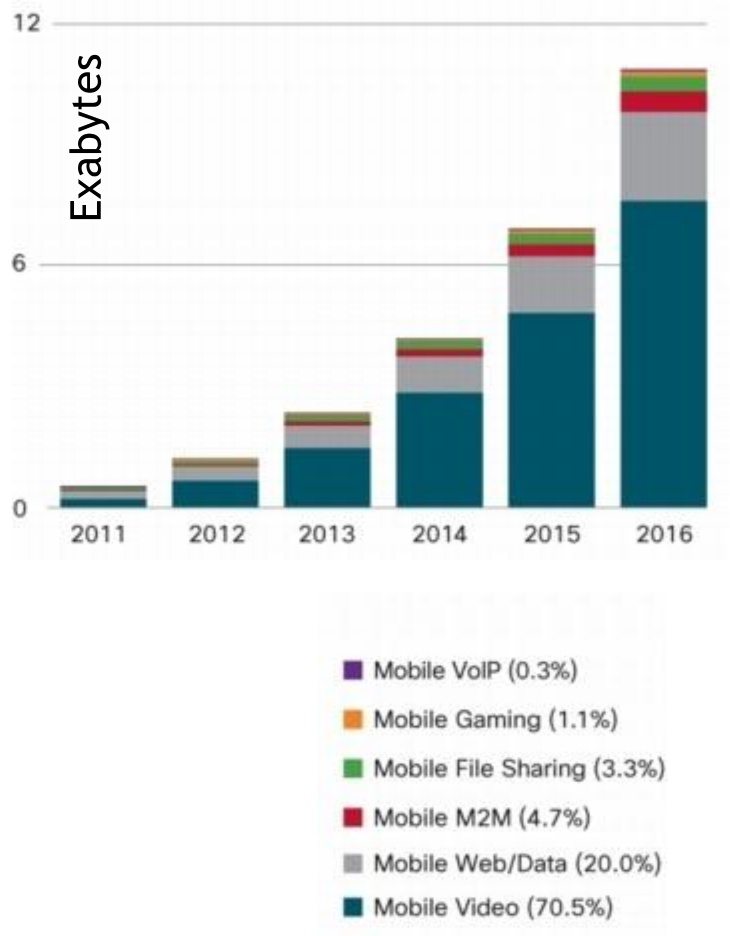


Sensor network

Motivation

- Mobile Video
  - 2011: 57% all mobile data traffic
  - 2016: More than 70%

- Smart Devices & Technology
  - iPhone4, iPad3 (retina display), Galaxy Note
  - 2011: 150 MB/mo → 2016: 2.6 GB/mo
  - LTE, LTE-Advanced, WiMAX
  - High spectral efficiency (UL/DL)



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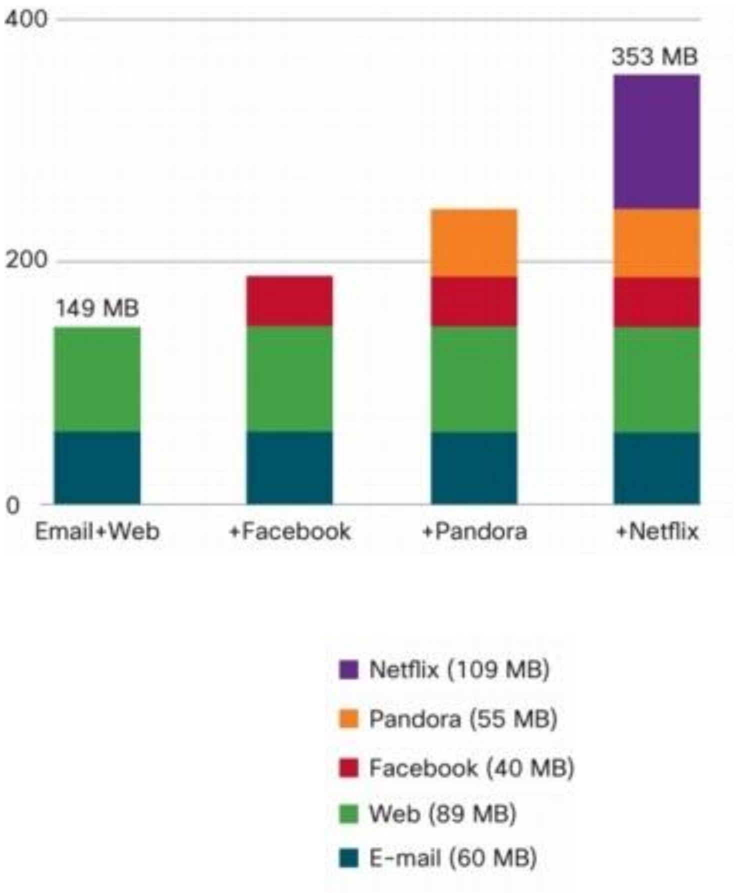
- Tiered Pricing
  - $10/GB beyond baseline quota
  - Throttling BW for top data users

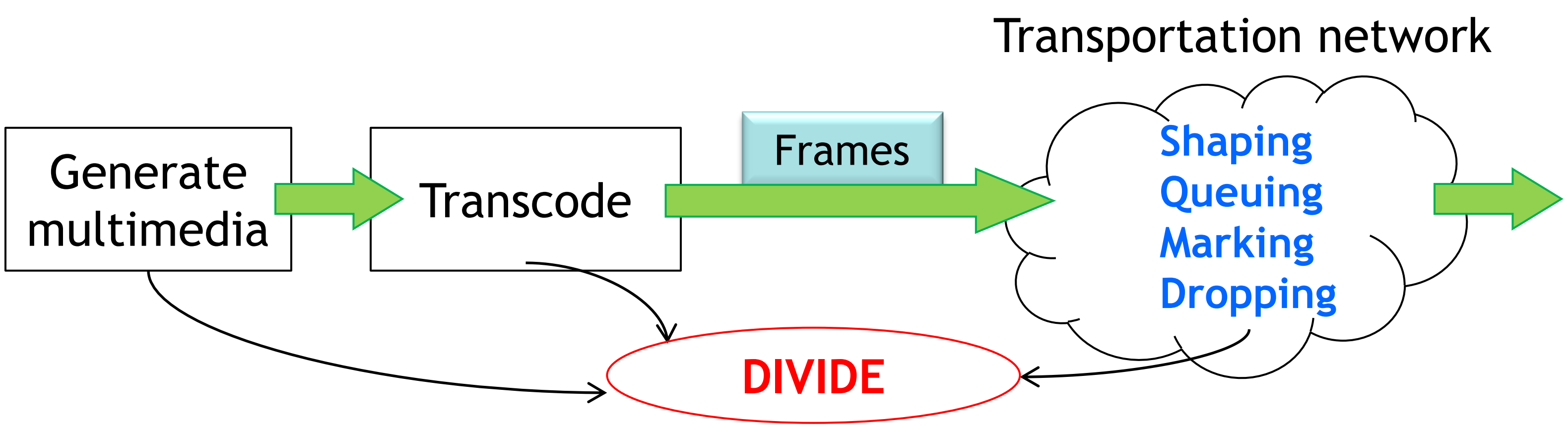


Congestion
Content-Pipe Divide

Content-Pipe Divide

- Content Providers
  - Media companies, end-users, operators of CDN and P2P systems
  - Generate content treating the network as just a means of transportation (dumb pipes)

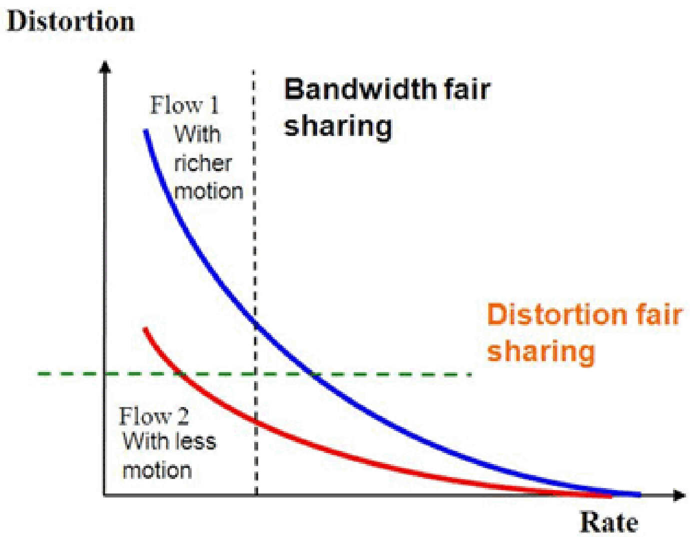
- Pipe Providers
  - ISPs, equipment and network management vendors, municipalities
  - Treat every content equally as just bits to transport between nodes



Two Sides of the Same Coin

- Content-Aware Network
  - Utilize content-characteristics to design RD-fair, adaptive protocols
  - Network resources allocated with optimality criteria reflective of content

Rate Fair: Each flow gets half the capacity
Rate-Distortion Fair: Flow1 gets more



- Network-Aware Content
  - Can the user consume more content without worrying about her wallet?
  - Is every bit needed for everyone at all times?
  - How can we generate network (quota)-aware content?

Content-Aware Optimization in W-CDMA (EV-DO, LTE)

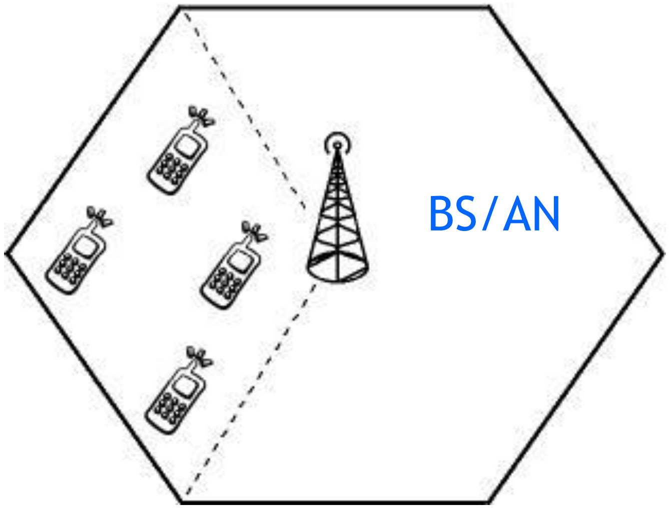
Focus: Connect the analytics of content-aware optimization with the specifics of engineering hooks and knobs in 3G/4G cellular standards (wrapper based)

Model

- Network: Cellular Uplink
  - Increasing demand for high data rate apps
    - EV-DO Rel 0 (153 Kbps)
    - EV-DO Rev A (1.8 Mbps)
    - LTE (50 Mbps)
  - Single hop uplink from MS to BS
  - Each user transmits a pre-encoded video upstream

- Degrees of Freedom
  - Scheduling
  - Transmit power control

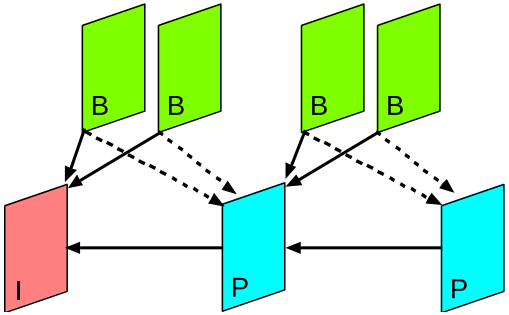
MS/AT



N : Number of MS

Model

- Video
  - Videos encoded as Group of Pictures (GOP)
  - GOP: sequence of frames, repeated over time
  - Dependent frames within one GOP; independent frames across GOPs



H.264 MPEG-4
GOP: IPBBPBB

Directed acyclic graph

- Frames
  - Length $\lambda(f_{ij})$
  - Decoder time stamp $t_{DTS}(f_{ij})$
  - Rate $r(f_{ij})$
  - Distortion $d(f'_{ij}, f_{ij})$

f_{ij} : Frame j of MS i

f'_{ij} : Reconstructed frame

Model

- Distortion Metric
  - PSNR (Peak Signal to Noise Ratio): spatial
  - MOS (Mean Opinion Score) 1 to 5

$$10\log_{10}\frac{255^2}{\sum_{i=0,j=0}^{w,h}(x_{ij}-y_{ij})^2}$$

$D_i(\Lambda_i)$: Distortion per GOP of MS i $\sum_{j=1}^{n} d(f'_{ij}, f_{ij}), \ \forall i$

Λ_i : Set of dropped frames per GOP by MS i

n : Number of frames per GOP

$R_i(\Lambda_i)$: Total rate to transmit one GOP by MS i $\sum_{j=1}^{n} r(f_{ij}) - \sum_{k=1}^{|\Lambda_i|} r(f_{il_k})$

Formulation

- GOP-Level CADF (Content-Aware Distortion-Fair) Optimization

Minimize the sum of distortions over a GOP for all video streams, subject to SINR and frame deadline constraints.

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^{N} D_i(\Lambda_i) \\ \text{subject to} & R_i(\Lambda_i) \leq \sum_{j=1}^{n} d \log(1 + c\gamma_{ij}), \quad \forall i \\ \text{variables} & 0 \leq p_{ij} \leq p_{max}, \quad \forall i, \quad \forall j \\ & \theta_{ij} \in \{0, 1\} \end{array}$$

NP-hard (MINLP)

where

$$\gamma_{ij} = \frac{p_{ij} g_{ii}(1-\theta_{ij})}{\sum_{i'=1, i' \neq i}^{N} p_{i'j'} g_{i'i}(1-\theta_{i'j'}) + \eta_0}$$

SINR

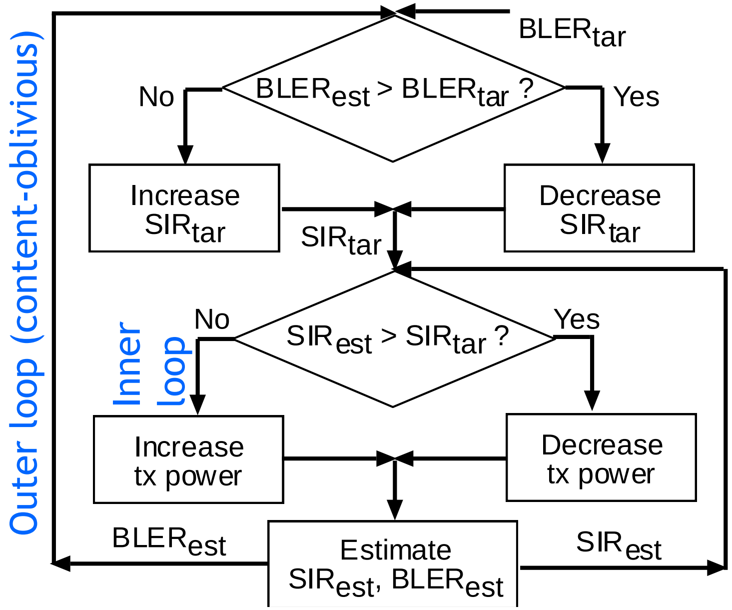
θ_{ij} : Scheduling decision variable – 0 if transmitted, 1 otherwise

p_{ij} : Transmit power for frame j of MS i

Power Control in W-CDMA

- Closed Loop Distributed Power Control
  (Foschini-Miljanic 1993)

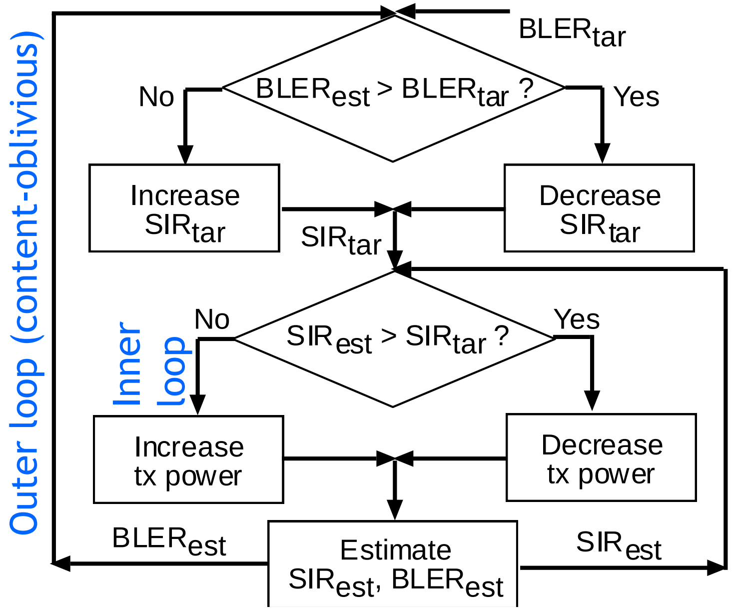
Converges to a feasible target SIR



Power Control in W-CDMA

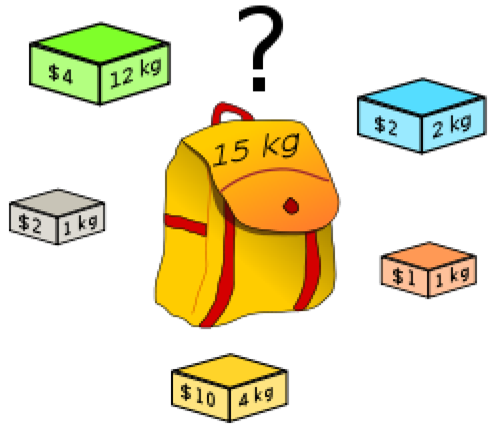
- Closed Loop Distributed Power Control
  (Foschini-Miljanic 1993)
  - Outer loop: BS estimates received BLER and selects a target SIR depending on whether the estimated BLER is above or below a target BLER (QoS class)
  - Inner loop: BS estimates received SIR and compares with target SIR to increase or decrease MS power

Converges to a feasible target SIR



Solving CADF Optimization

- Multiple Knapsack Problem (MKP) Mapping
  - Generalization of the single knapsack problem
  - Items = frames
  - Profit = negative frame distortion
  - Weight = frame rate
  - Knapsack capacity = channel capacity



- Key Difference
  - Knapsack capacities are not interchangeable
    - A restricted MKP
  - Knapsack capacities are interdependent
    - Signal of MS i is interference to all other MS j

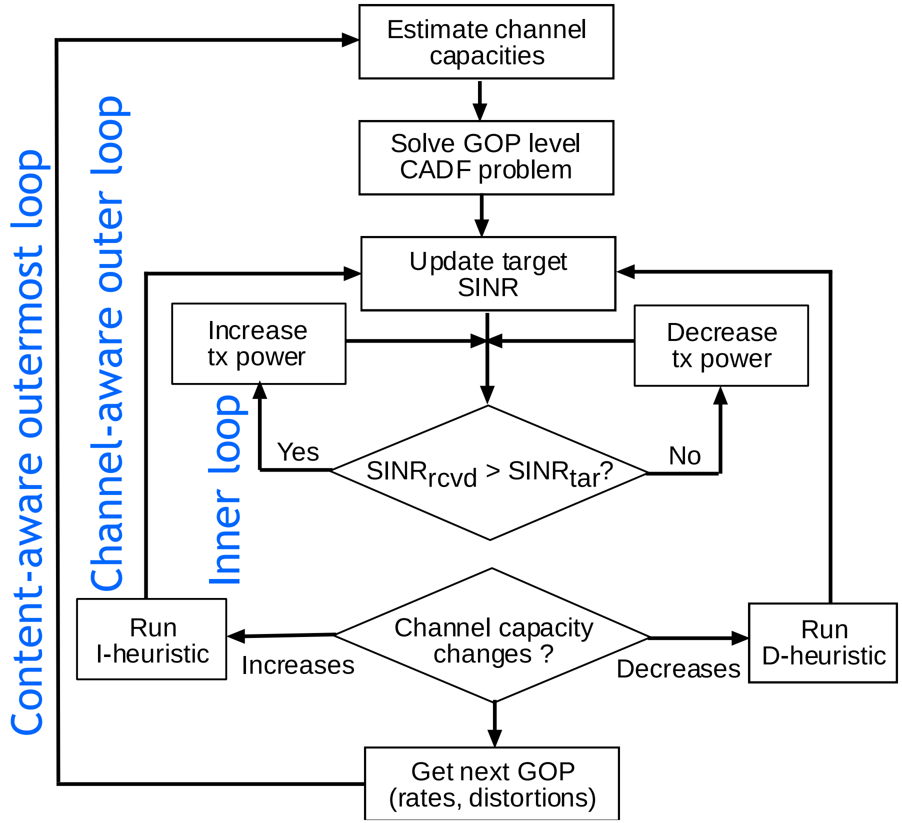
PTAS for MKP executed multiple times
+
Two heuristics to incorporate channel variation

RD-Fair Power Control in W-CDMA

- Content-Aware Outermost Loop

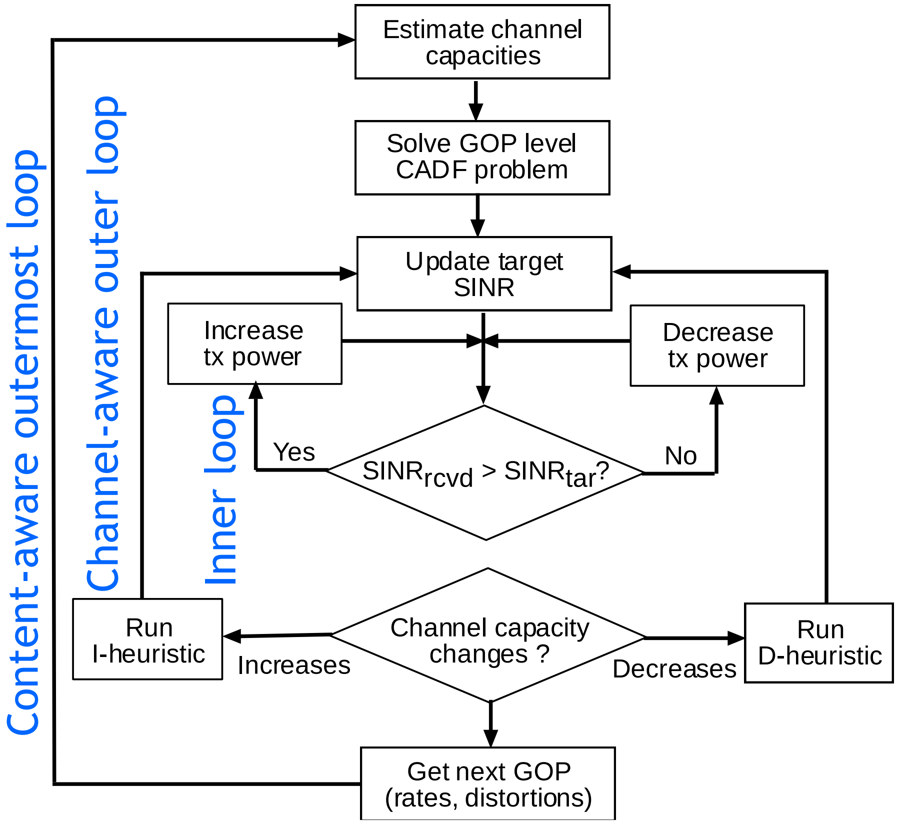
- Channel-Aware Outer Loop

- Inner Loop



RD-Fair Power Control in W-CDMA

- Content-Aware Outermost Loop
  - Executes in the beginning of every GOP (runs PTAS multiple times)
  - Adjusts a target SINR for each MS: determines optimal set of frames to transmit and powers
- Channel-Aware Outer Loop
- Inner Loop



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- Content-Aware Outermost Loop
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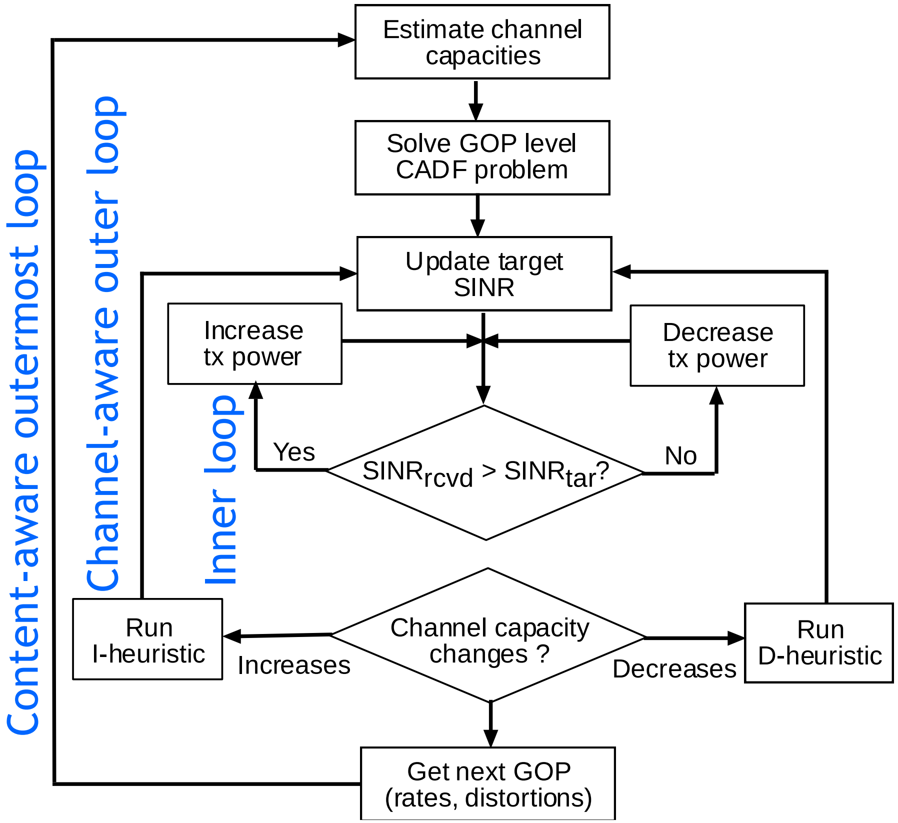
- Channel-Aware Outer Loop
  - Once every GOP to detect any channel variation during execution of the outermost loop
  - I-heuristic (increase in capacity)
  - D-heuristic (decrease in capacity)
- Inner Loop



RD-Fair Power Control in W-CDMA

- Content-Aware Outermost Loop
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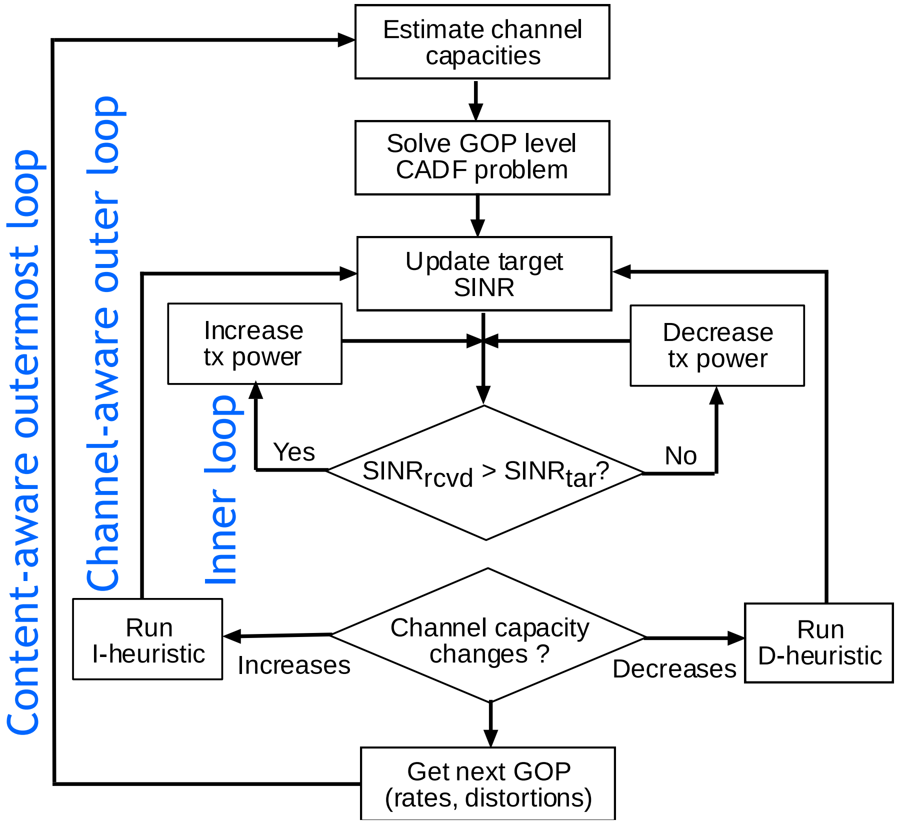
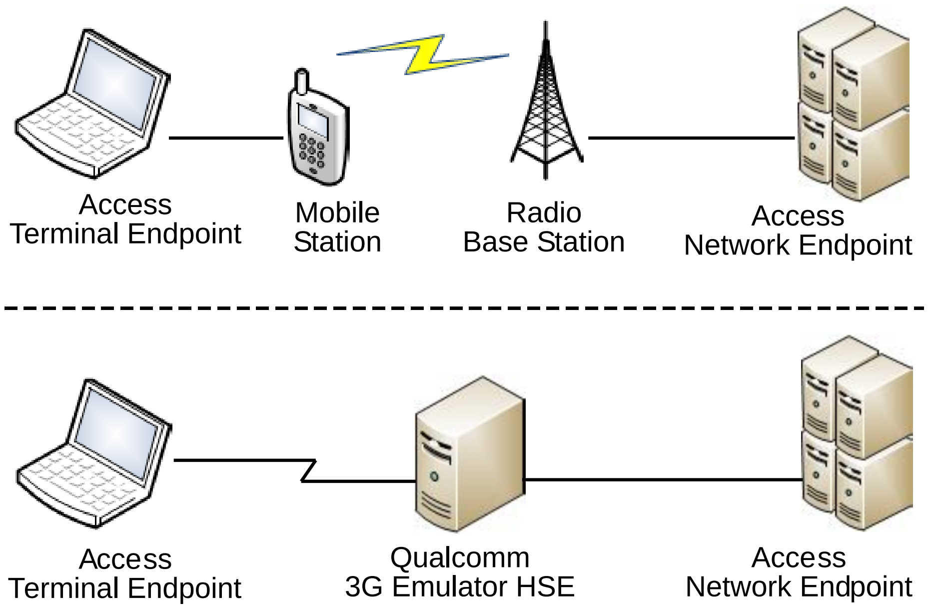
- Channel-Aware Outer Loop
  - Once every GOP to detect any channel variation during execution of the outermost loop
  - I-heuristic (increase in capacity)
  - D-heuristic (decrease in capacity)

- Inner Loop
  - Adjusts transmit power depending on received SINR (same as Foschini-Miljanic)

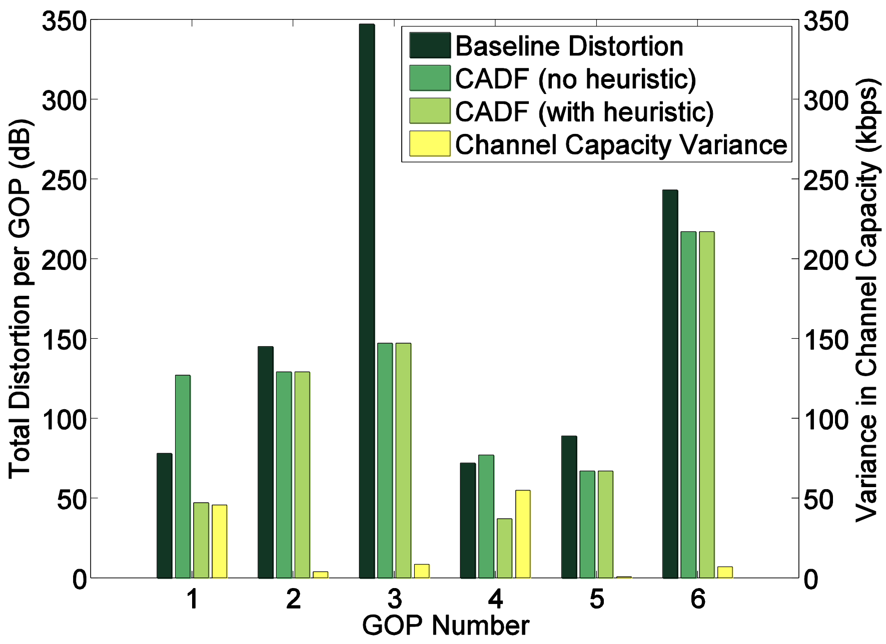


Evaluation of CADF

- Testbed: Qualcomm High data rate System Emulator (HSE)
- Videos
  - 5 laptops as MS (vehicular antennae)
  - Each video: 12 sec, 250 frames, 6 GOPs (~500 slots)
  - Reverse Activity Bit (RAB) set to high to simulate heavy non-video traffic
  - Each channel profile comprises 20,000 slots (2 ms each)

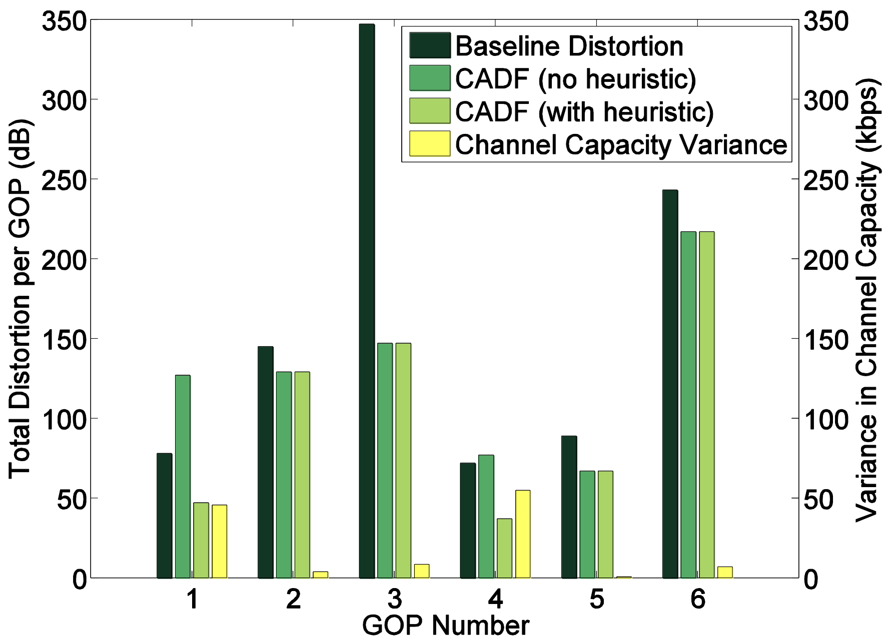


Evaluation of CADF



GOP-level distortion: Comparison of CADF with Foschini-Miljanic

Evaluation of CADF



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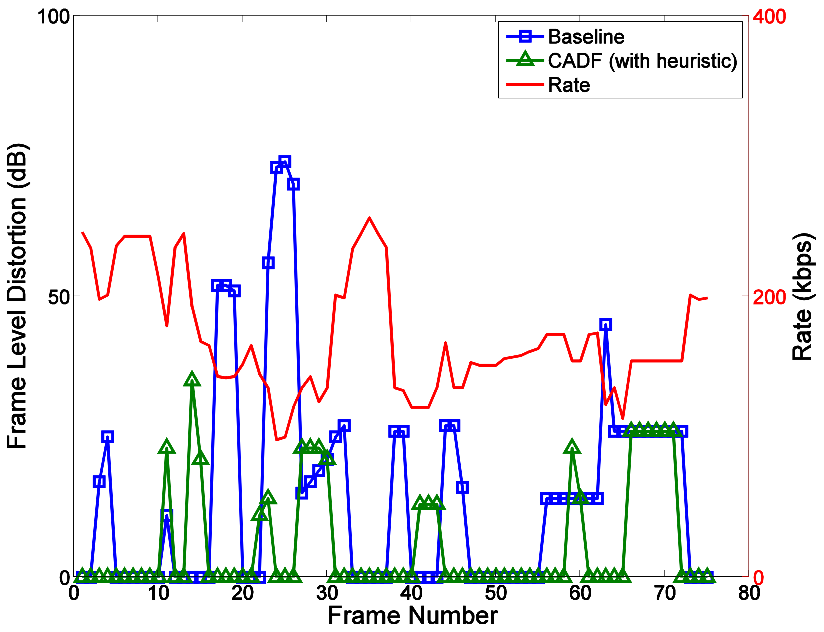


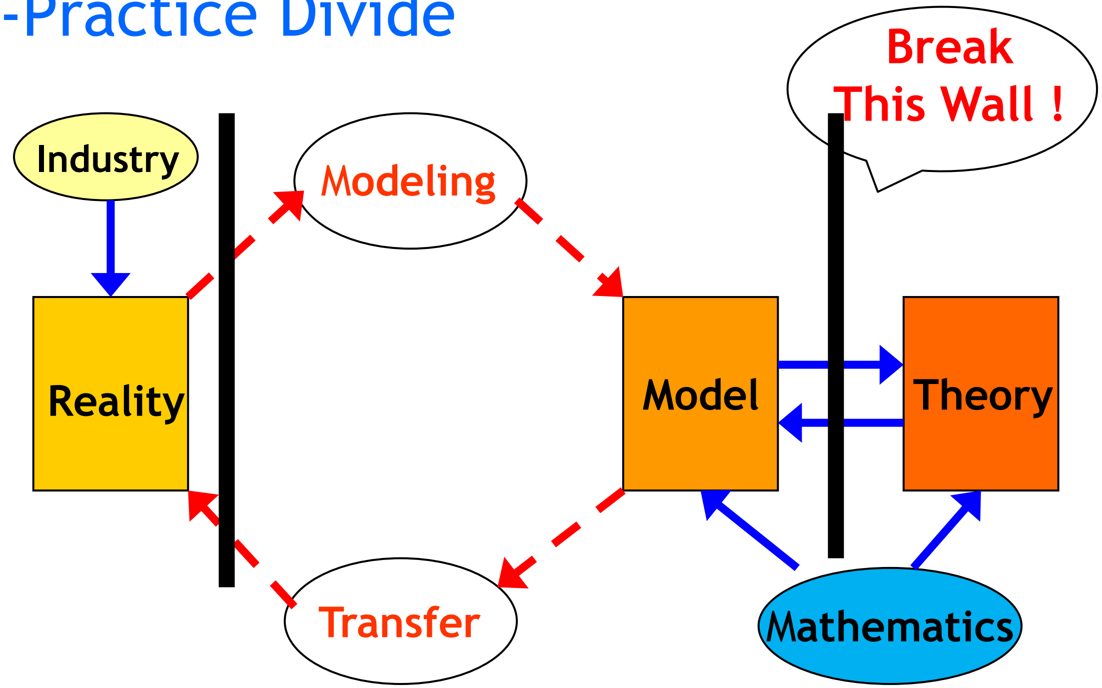
Frame-level distortion: Comparison of CADF with Foschini-Miljanic

Implementation

Transmit power control on WARP software defined radio

Implementation

- Theory-Practice Divide



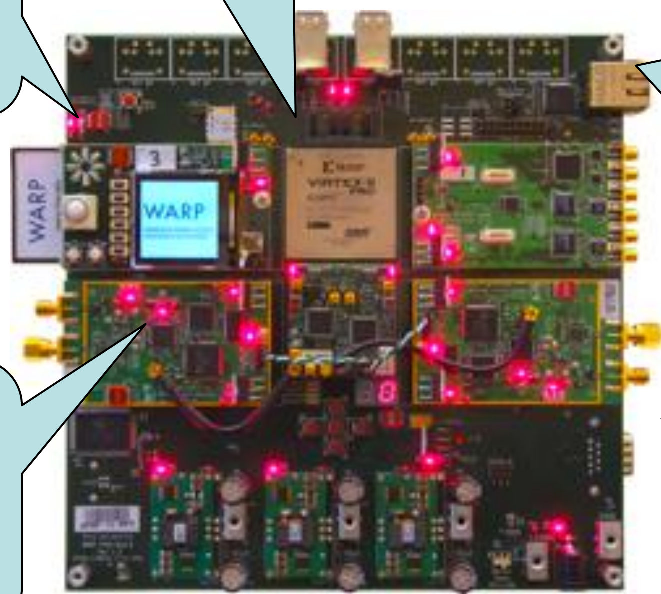
WARP

Wireless open-Access Research Platform (Rice)

Programmable through memory card or USB connection

Xilinx FPGA based

Ethernet connection to host computer for data exchange



Radio board for unlicensed band with two antenna connections

Manual control provided through the serial port

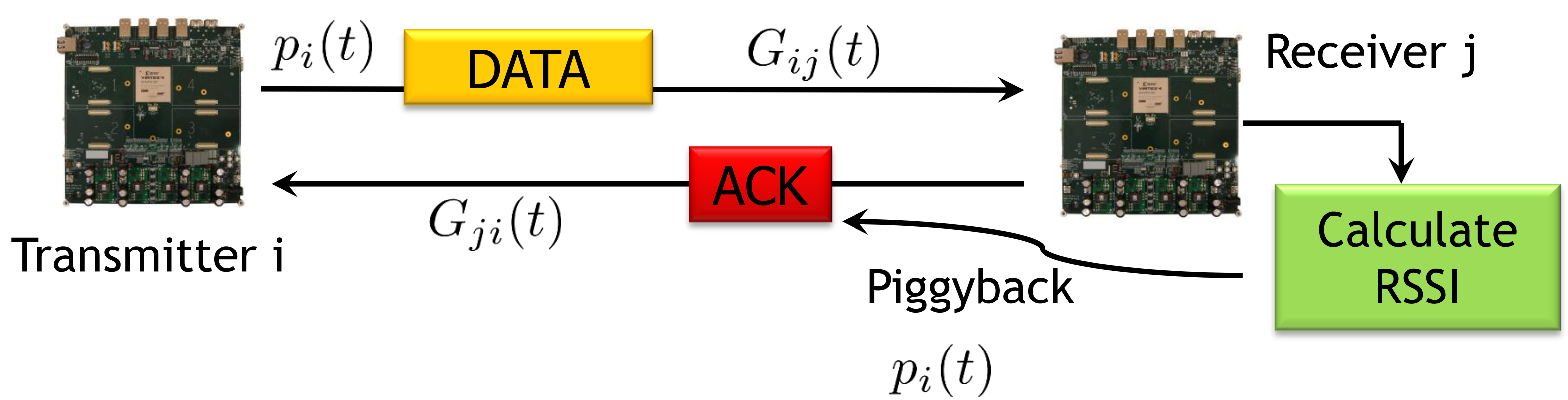
WARP

- PHY: 802.11 OFDM
  - Carriers
    - 64 carriers across 10 MHz, 802.11p (vehicular) like signal
  - Transmit Power
    - Adjustable at 0.5 dB (-20 dBm to 10 dBm)
  - Modulation
    - BPSK, QPSK, 16-QAM, 64-QAM

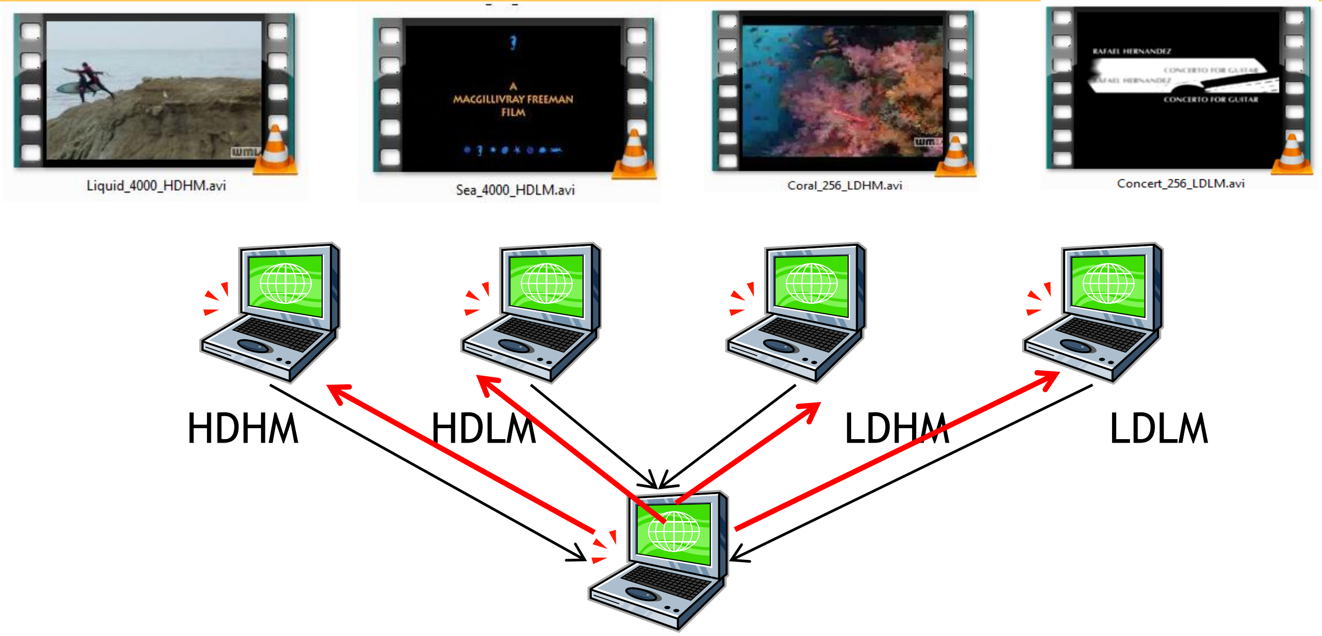
- MAC: 802.11 DCF
  - Carrier sensing through energy detection
  - Exponential random back-off (could be disabled)
  - Retransmission is packet is not acknowledged

RSSI as a Proxy for SINR

- Received Signal Strength Intensity
  - Average of 16 received signal samples
  - Taken after the low-noise amplifier
  - Varies with
    - TX power, channel gain, interference
    - Mobility
    - Automatic Gain Control (AGC) adjustments (1 to 2 dB)



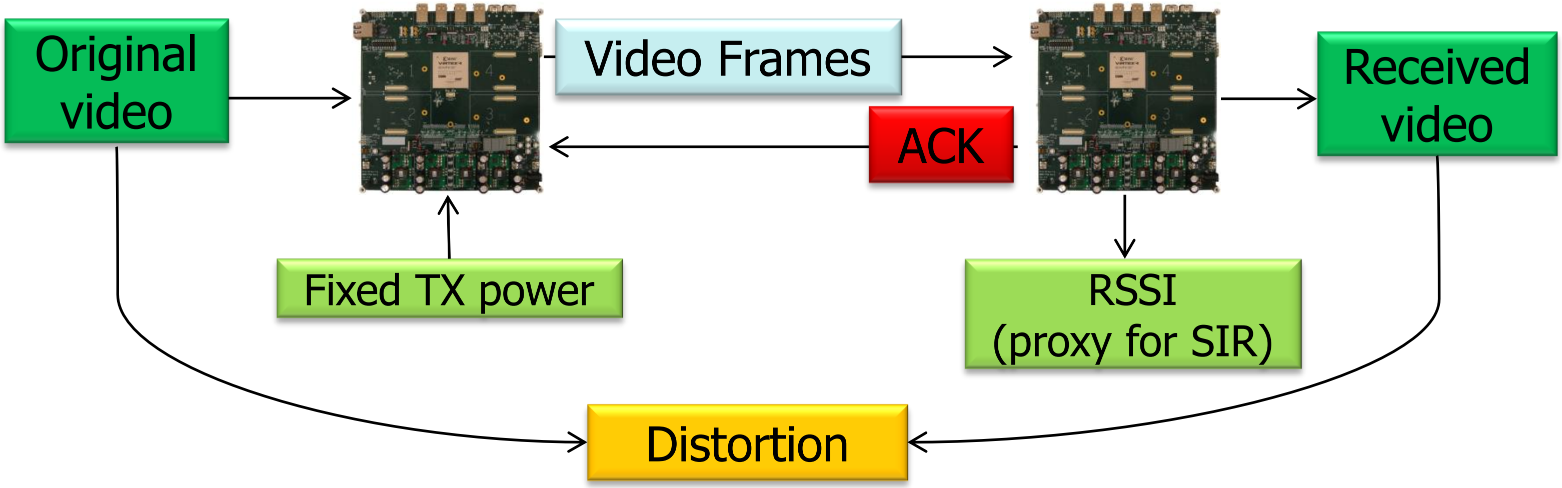
Experimental Setup



- High-definition
  - Video: 4000 kbps, MPEG-4 v3, 29.97 fps, 1280 x 720
  - Audio: 44.1 kHz, 128 kbps, MPEG-1 L3
- Low-definition
  - Video: 256 kbps, MPEG-4 v3, 29.97 fps, 640 x 320

Video Profiling

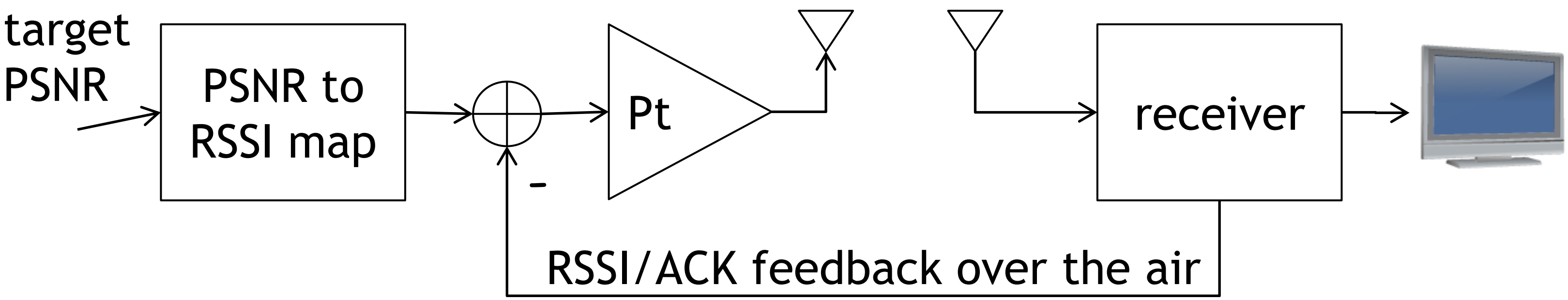
- Goal
  - For each video, create a table that relates TX power => RSSI => Distortion



- Procedure
  - Connect one WARP transmitter to a WARP receiver with a cable
  - For TX power = -20 dBm to 10 dBm in steps of 2 dBm
    - Send and save video on the receiver
    - Calculate distortion (frame-by-frame) of the received video

Closed Loop Power Control

- Change power based on signal strength difference
  - If out of range tough luck: set max or min
  - Can be done on a per packet basis (fast scale)



- Algorithm
  - Measure RSSI from ACK, and lookup corresponding distortion from table
  - Lookup target RSSI and target PSNR from table
  - If difference in PSNR is more than x
  - Adjust power level by:

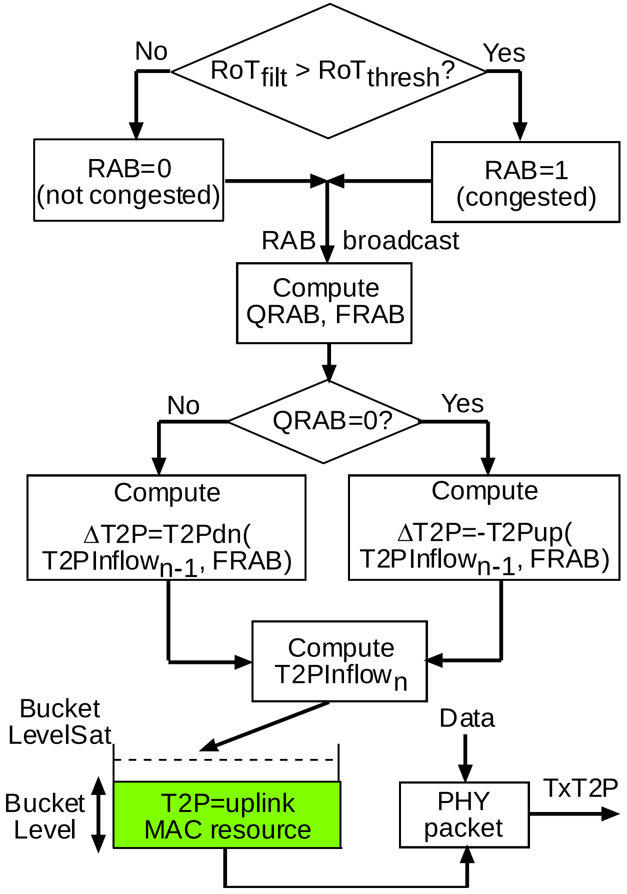
$$\Delta P_t = RSSI_{target} - RSSI_{\mathrm{measured}}$$

YouTube Demo Video

[YouTube demo video on power control]

Ongoing: EV-DO Rev A

- Coordinated Power Control
  - AN controls RoT by setting RAB and does long term power allocation (T2PInflow)
  - AT decides per-PHY packet power allocation (TxT2P) from T2PInflow using a token bucket



Ongoing: EV-DO Rev A

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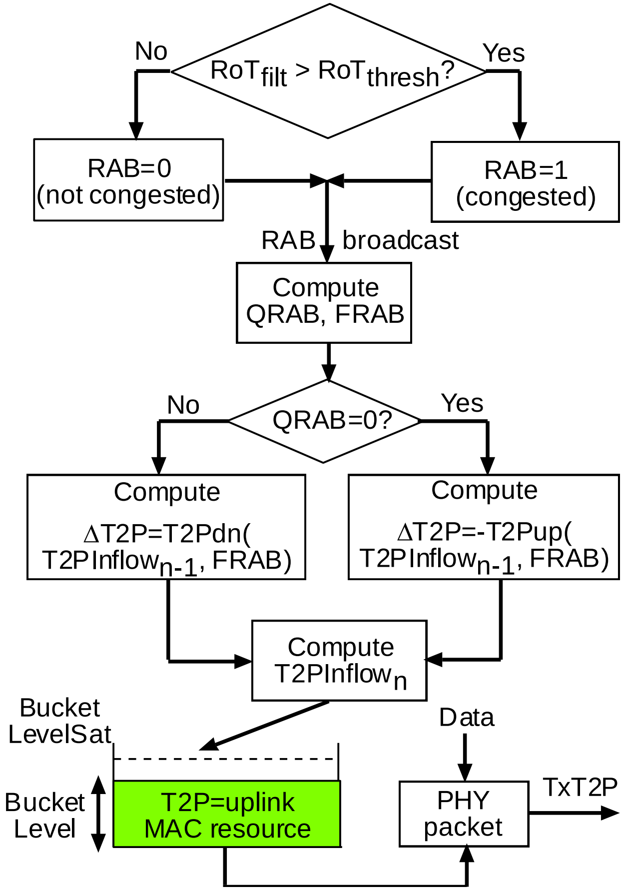
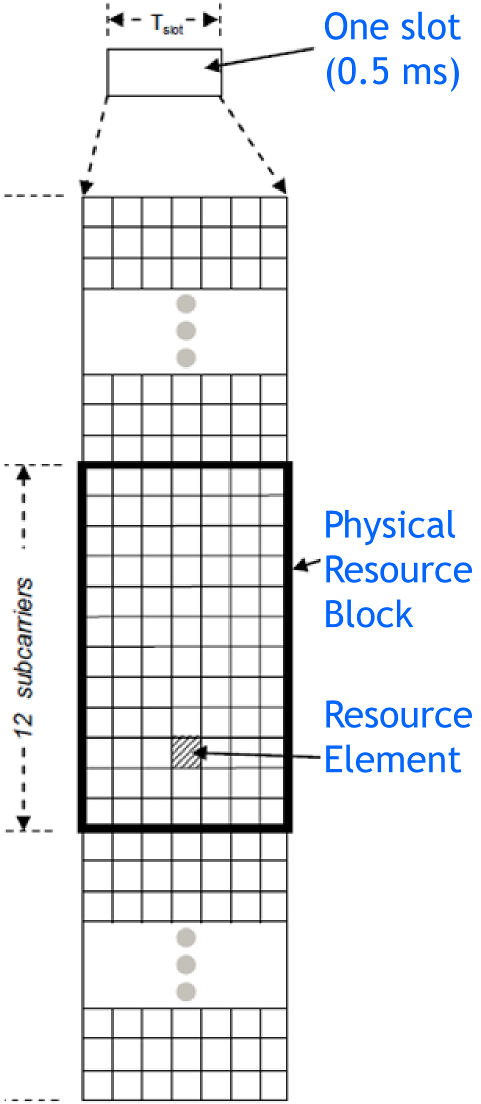
- Content-Aware Token Bucket
  - Key Idea: Accumulated T2PInflow at the end of GOP g is used for allocation in GOP g+1
  - Similar constrained optimization formulation as in W-CDMA
- Content-Aware T2PInflow Allocation



Ongoing: LTE

- Ideas
  - Make the allocation of the PRBs to the UEs content-aware



QAVA: Quota-Aware Video Adaptation

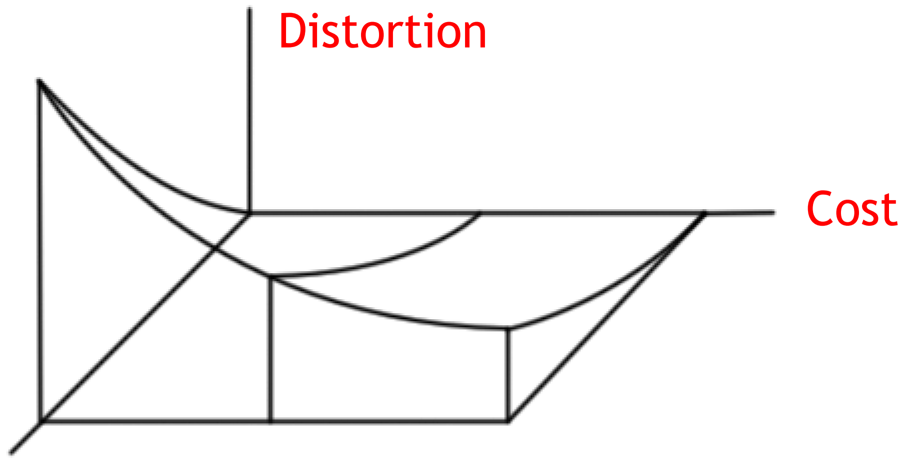
Video traffic becoming dominant

vs.

Usage-based pricing becoming prevalent

- How can the user consume more content without worrying about her wallet?
- Is every bit needed for every user at all times?
- How can we generate (network) quota-aware content?

The Solution: QAVA

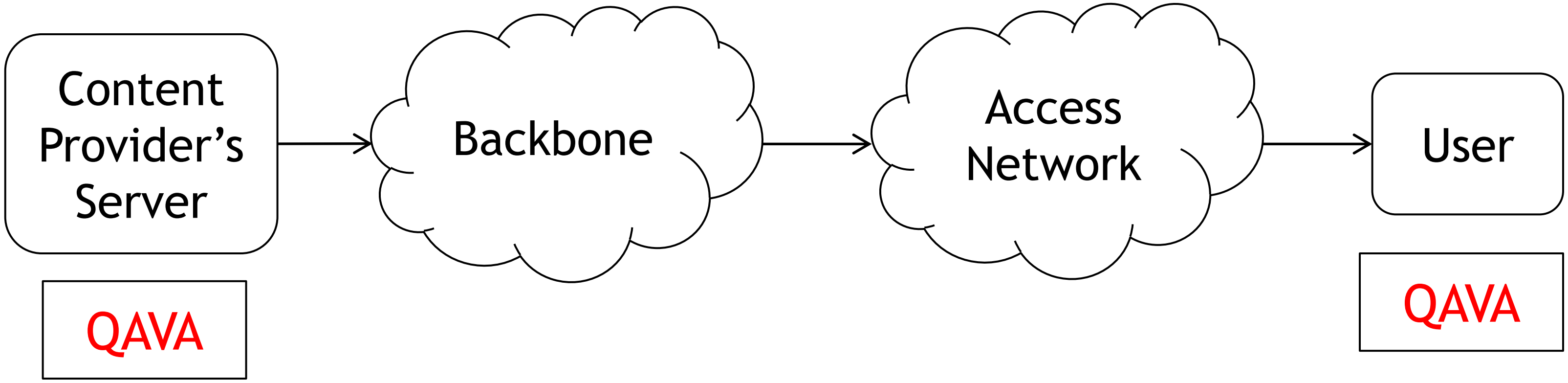


- Three Goals
  - Cost minimization
  - Video distortion minimization
  - Number of videos watched

- QAVA provides a graceful, tunable tradeoff

QAVA System Design

- Adaptively choose appropriate bitrates
  - Video Profiler (VP): Leverage video compressibility from motion vectors
  - User Profiler (UP): Predict user's future data consumption from past history
  - Stream Selector (SS): Choose appropriate bitrate to maximize video quality subject to budget



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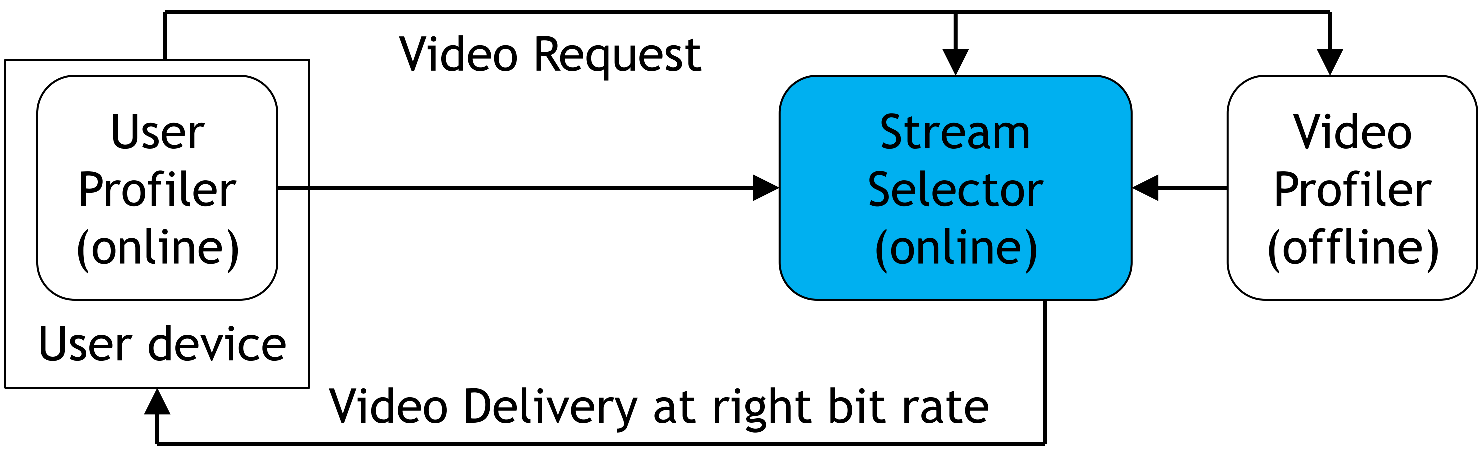

Stream Selection without UP

- Notations

  B : Total budget

  N : Total videos requested

  M_i : Number of versions of video i

  u_{ij} : Utility of version j of video i

  c_{ij} : Cost of version j of video i

  x_{ij} : 1 if version j of video j is selected; 0 otherwise

- Formulation

  Maximize the sum of utilities of all the selected versions of the N requests, subject to

  - ❑ Exactly one version of each request is granted
  - ❑ Total cost of all the selected versions must be within budget

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$$\underset{x_{ij}\in\{0,1\}}{\text{maximize}} \quad \sum_{i=1}^{N}\sum_{j=1}^{M_i} u_{ij}x_{ij}$$

$$\text{subject to} \quad \sum_{j=1}^{M_i} x_{ij} = 1, \quad \forall i$$

$$\sum_{i=1}^{N}\sum_{j=1}^{M_i} c_{ij}x_{ij} \leq B,$$

Online Multi-Choice Knapsack Problem

- Formulation

  Maximize the sum of utilities of all the selected versions of the N requests, subject to

  - Exactly one version of each request is granted
  - Total cost of all the selected versions must be within budget

Algorithm

- Overall Idea (Modified Online MCKP)
  - Transformation: Subtract the utility and cost of the cheapest stream from each video

    New utility, cost, budget: u'_{ij} c'_{ij} B'
  - Assume bounds of utility to cost ratio are known (by training, in practice)

    $$L' \leq u'_{ij}/c'_{ij} \leq U'$$
  - Define efficiency threshold:
    b(t) = budget consumed until t

    $$\Phi(b(t), B') = \left(\frac{U'e}{L'}\right)^{b(t)/B'} \cdot \left(\frac{L'}{e}\right)$$
  - For video request i at time t:
    streams with better utility to cost ratio than efficiency threshold

    $$\mathcal{S}(t) := \{j \mid u'_{ij}/c'_{ij} \geq \Phi(b(t), B')\}$$
  - Choose stream j* that has max utility in S(t) and fits within budget
    Competitive ratio: $\log(U'/L') + 1$

Stream Selection with UP

- Notations

  T : Length of a billing cycle

  P : Total periods of equal length in T

  A_p : Set of videos requested in period p

  B_p : Budget for period p

$$
\begin{aligned}
\underset{x_{ij}\in\{0,1\}}{\text{maximize}} \quad & \sum_{i\in A_p}\sum_{j=1}^{M_i} u_{ij}x_{ij} \\
\text{subject to} \quad & \sum_{j=1}^{M_i} x_{ij} = 1, \quad \forall i \in A_p \\
& \sum_{i\in A_p}\sum_{j=1}^{M_i} c_{ij}x_{ij} \le B_p,
\end{aligned}
$$

Multiple Time-Sequential Knapsacks

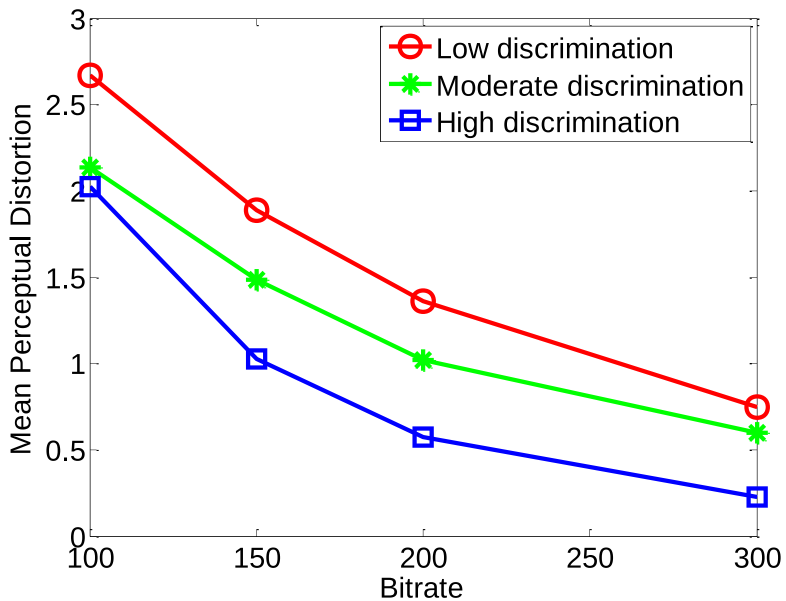
- User Profiler
  - Predicts a user's data consumption for period p while at period p-1
  - Online triple exponential smoothing (modified Holt-Winters): Online-TES
    - Seasonality
    - Trend

Evaluation of QAVA

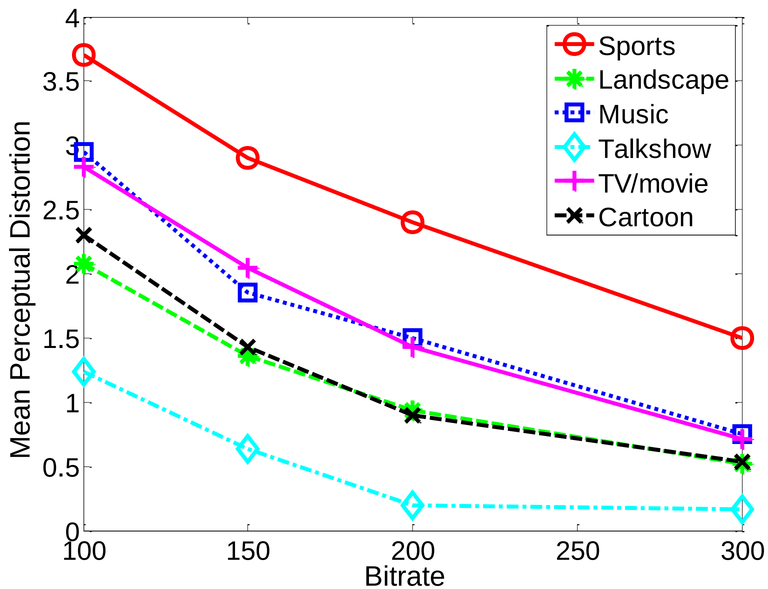
- Human Subject Study 1
  - To determine subjective video quality (MOS utility) and build Video Profiler
  - Twenty diverse H.264 clips of resolution 640 x 480 and duration 20 sec
  - Each video encoded at 100, 150, 200, 300, Kbps
  - Videos shown to 20 participants on iPhone4 held ~50 cm from their eyes
  - Participants rated in the scale 1 (imperceptible) to 5 (very annoying)

Evaluation of QAVA

- Human Subject Study 1



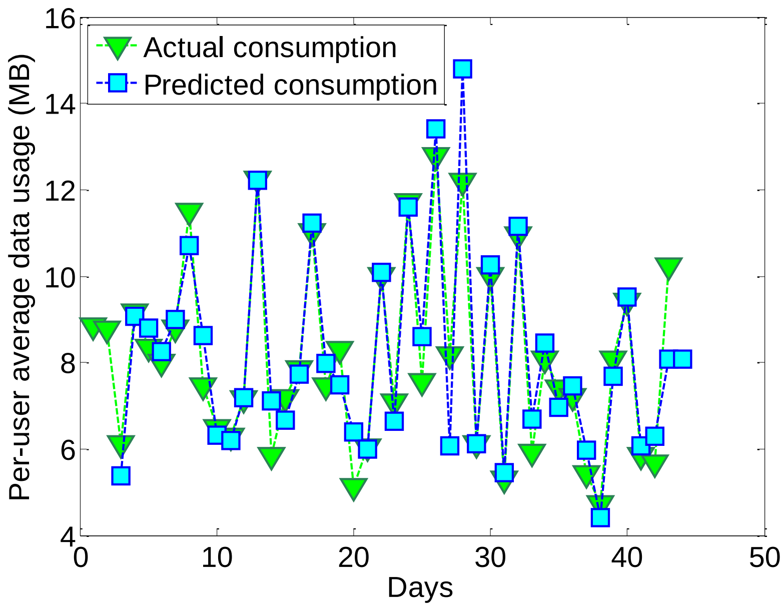
Rate-distortion curves for different types of users, averaged across all videos



Rate-distortion curves for different types of videos, averaged across all users

Evaluation of QAVA

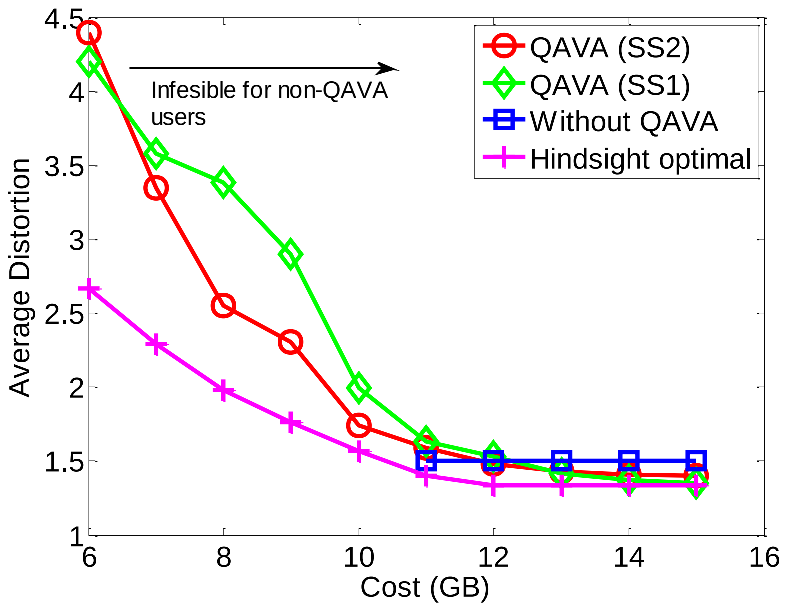
- Human Subject Study 2
  - To evaluate the User Profiler (Online-TES)
  - Mobile data usage from 30 iPhone and 20 iPad users at Princeton
  - Collected every 3 hours for 43 days



Training data set: days 1-20
Test data set: days 21-43
Average fractional error = 6.9%

Evaluation of QAVA

- Overall System Simulation
  - 1430 video requests randomly generated over 30 days (average time spent by a Netflix user)
  - Video duration normally distributed with mean 30 sec and s.d. 5 sec



Cost-distortion tradeoff: For a fixed set of videos, a non-QAVA user cannot watch all videos below a 11 GB quota

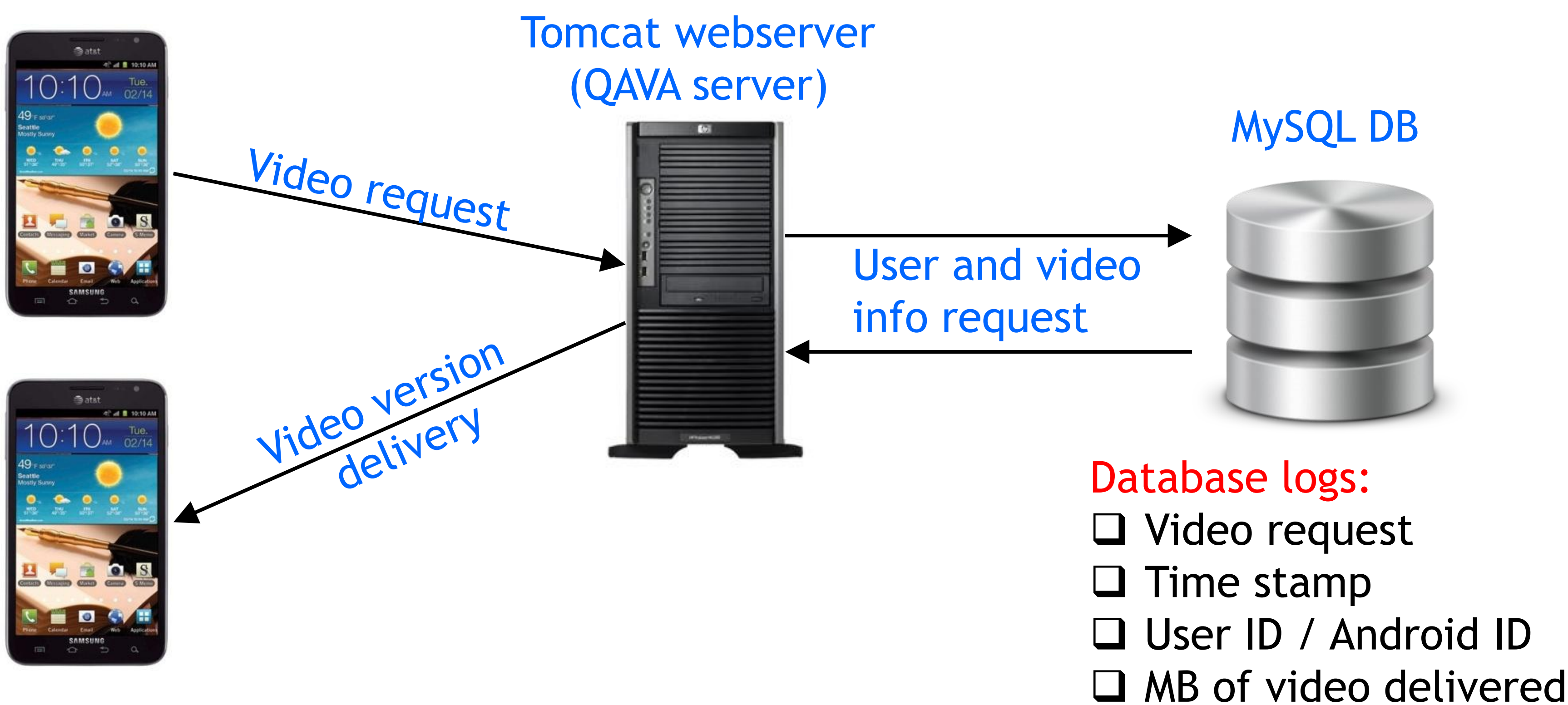
Hindsight optimal: benchmark with perfect knowledge of all future requests

Implementation on Android in Java

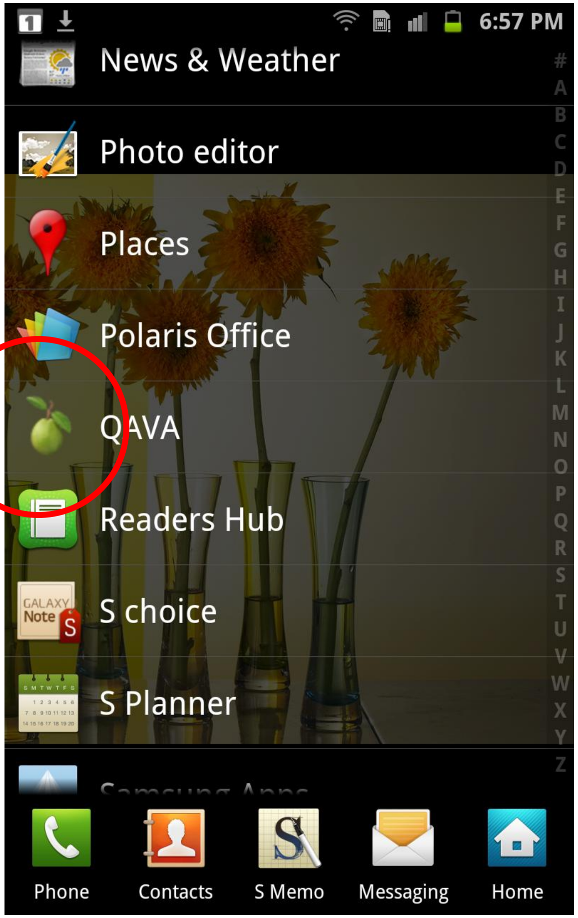
(earlier on Silverlight in C#)

Princeton Trial

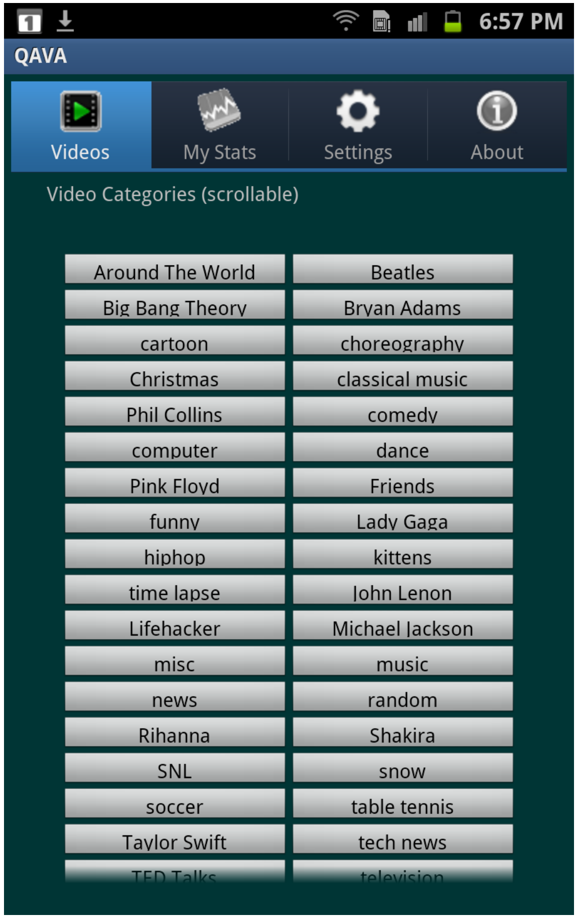
- Set Up
  - 25 volunteers with Android phones
  - ~500 videos encoded at 25 Kbps granularity (100 Kbps – 500 Kbps)



Android App Screenshots



Android App Screenshots



Android App Screenshots

Summary of Other Projects

- Multi-class traffic management in data center backbone networks
- Throughput-delay tradeoffs for fast data collection in sensor networks

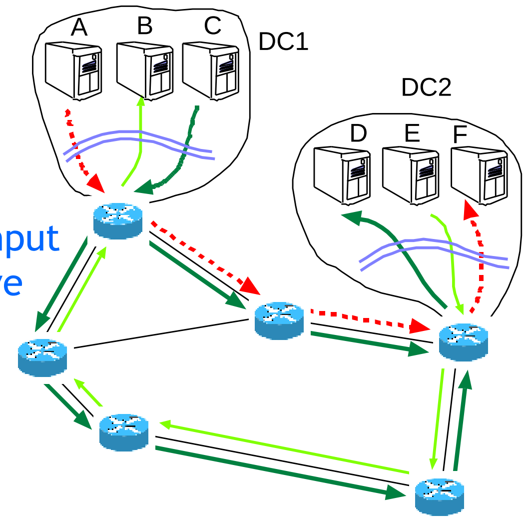
Datacenter Backbone Traffic Management

- **Inter-Datacenter Resource Allocation**

  Scalability

  - Multiple classes with performance requirements
  - Sessions in each class with priorities
  - Multi-path routing
  - Tradeoff between optimality, message-passing, and convergence

delay-throughput Sensitive traffic

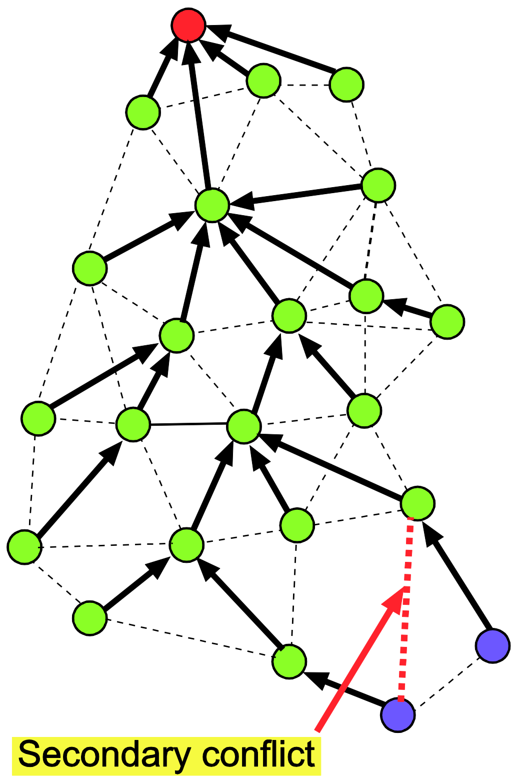


- **Network Utility Maximization Problem**
  - Design a scalable network architecture and distributed traffic engineering protocol maximizing a weighted sum of utilities of all the sessions across all classes and datacenters

- **Contributions**
  - Primal and dual decomposition-based optimization
  - Semi-decentralized, tiered architecture with distributed functionality across tiers

Throughput-Delay Tradeoff in Sensornets

- Maximize Throughput & Minimize Delay
  - Joint frequency and time slot scheduling using multiple channels
  - Graph and SINR interference model
  - Multiple orthogonal frequencies, TDMA scheduling
- Contributions
  - Approximation algorithms with constant factor (UDG) and logarithmic (RGG) guarantee
  - Bi-criteria approximation algorithms with constant factor guarantee for optimal routing topologies
  - Implementation on Tmote Sky motes (802.15.4)



Conclusions

- Content-Aware Distortion-Fair (CADF) optimization for video delivery in cellular networks
- Online quota-aware video adaptation (QAVA)
- Comprehensive architecture and algorithms with three modules: VP, UP, SS
- Implementation on WARP software defined radios and Android devices (ongoing trial in Princeton)

- Roadmap (Networking)
  - Scalability
  - Heterogeneity (sensors, femto, Wi-Fi, cellular)
  - Content-Awareness
  - Cooperative relays, BS, users
  - Converged Network of Networks
- 4G LTE
  - eNB coordinated transmission (macro diversity)
  - Content-aware LTE and LTE-A
  - Mobility (Fuzzy cells)
  - Bandwidth aggregation from heterogeneous devices

Demo Video

YouTube demo video on QAVA

Model

- Distortion Metric
  - Capturing activity